

Swiss Institute of Banking and Finance



University of St.Gallen

Case Study Session 2: Valuing Snap After the IPO Quiet Period

3,130/4,130 Corporate Finance

Key messages of this case study / exercise

1. Understand the DCF valuation issues and illustrate the challenges of valuing young, rapidly growing technology firms using DCF.
2. Understand the role research analysts play in conveying value-relevant information as well as the potential conflicts of interest faced by IPO-affiliated sell-side analysts.
3. Analyze the publication effect of estimates based on DCF model that contained an error and a revised version that was published the next day on stock prices.
4. Explore the process of going public and two anomalies associated with IPOs: short-term underpricing and long-term underperformance.

Outline

1. Recap: DCF model and cost of capital
2. IPO: The process for the Snap company
3. Kemp's challenge: Elisabeth Kemp's challenge on the day the IPO quiet period expired
4. Valuation of Snap: Morgan Stanley's DCF model and assumptions
5. Conflicts of interest: Affiliated analysts' recommendations
6. Excursus: Publication of Morgan Stanley's revised research notes

1. Recap: DCF model – Free cash flows as input

- To calculate the free cash flow to the firm, which could be potentially distributed to a firm's creditors and shareholders, the cash flow has to be adjusted for reinvestment needs:

	EBIT	Earnings before interest and taxes
	- Taxes	Taxes on EBIT
Adjustments for Reinvestments	NOPAT	Net operating income after taxes
	± non-cash items	e.g., adding depreciations, deduction of non-cash valuing gains
	-CAPEX	Capital expenditures
	-ΔWC	Change in non-cash working capital
	FCF	Free cash flow to the firm

- Any tax benefits from interest expenses are neglected.

1. Recap: Estimating horizon values

- The final payment at H is often called **terminal or horizon value**.
- To determine the **horizon value** PV_H , it is often assumed that the free cash flows at the end of the detailed planning period grow uniformly at a constant rate g .
- Hence, the horizon value can be determined as:

$$\begin{aligned} PV_H &= \frac{FCF_{H+1}}{1+r} + \frac{FCF_{H+2}}{(1+r)^2} + \frac{FCF_{H+3}}{(1+r)^3} + \dots = FCF_{H+1} \cdot \left(\frac{1}{1+r} + \frac{(1+g)}{(1+r)^2} + \frac{(1+g)^2}{(1+r)^3} + \dots \right) \\ &= \frac{FCF_{H+1}}{r-g} \end{aligned}$$

- Thus the present value of a corporation is:

$$PV_0 = \frac{FCF_1}{1+r} + \frac{FCF_2}{(1+r)^2} + \dots + \frac{FCF_H}{(1+r)^H} + \frac{1}{(1+r)^H} \diamond \frac{FCF_{H+1}}{r-g}$$

1. Recap: Company cost of capital – WACC

- Definition: The discount rate for the firm's **average-risk** projects.
- The company's cost of capital can be understood as the expected return on a portfolio of **all** the company's existing securities.
 - This portfolio usually includes both debt and equity. It is also referred to as Weighted Average Cost of Capital (WACC).

$$WACC = r_A = \frac{E}{D + E} r_E + \frac{D}{D + E} r_D$$

- E : Market value of the equity (Number of shares · Price per share)
- D : Market value of the debt (often approximated by book value)
- r_E : Cost of equity
- r_D : Cost of debt

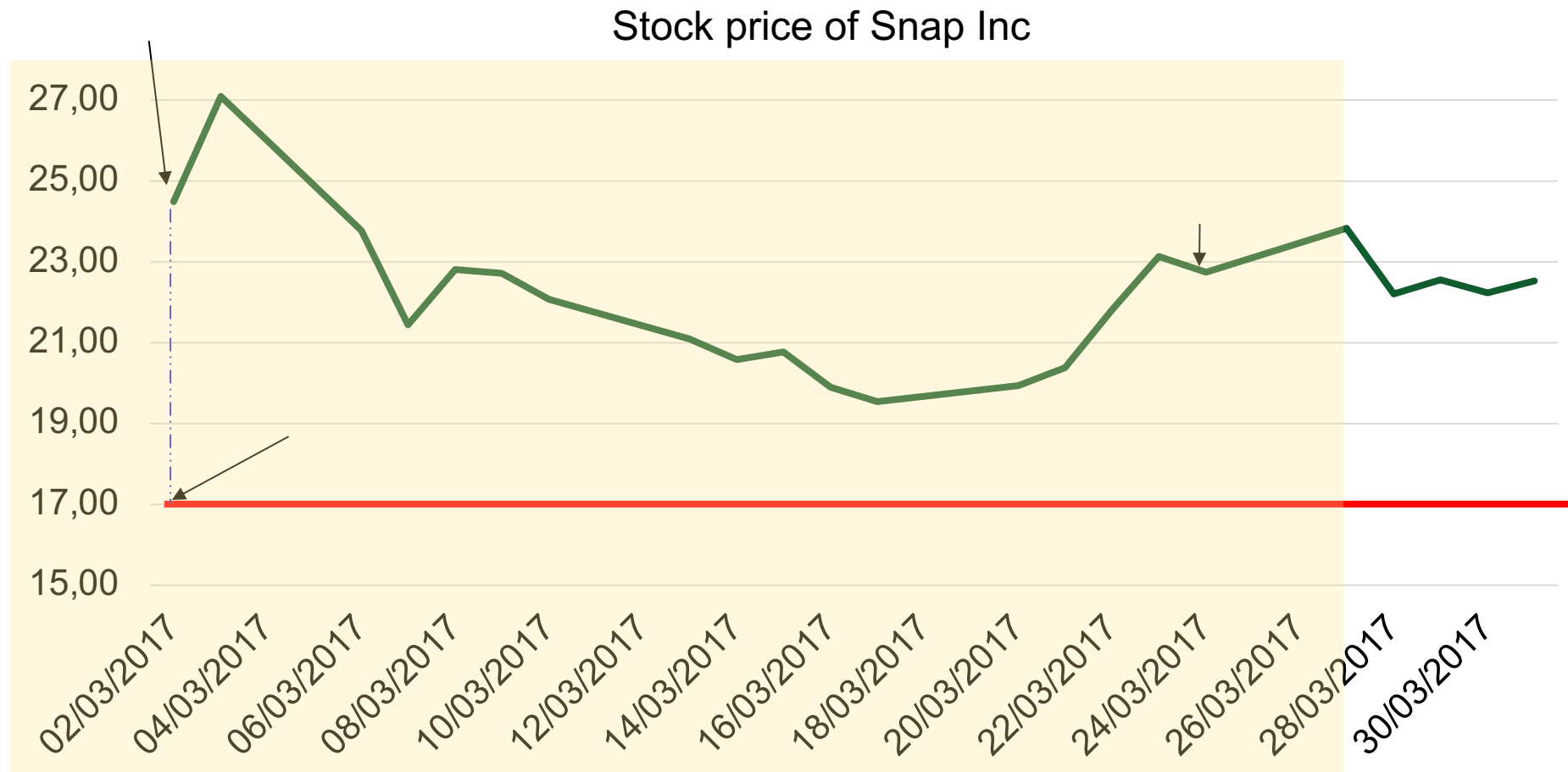
2. IPO: How did the company evolve over time?

Picaboo

Snapchat

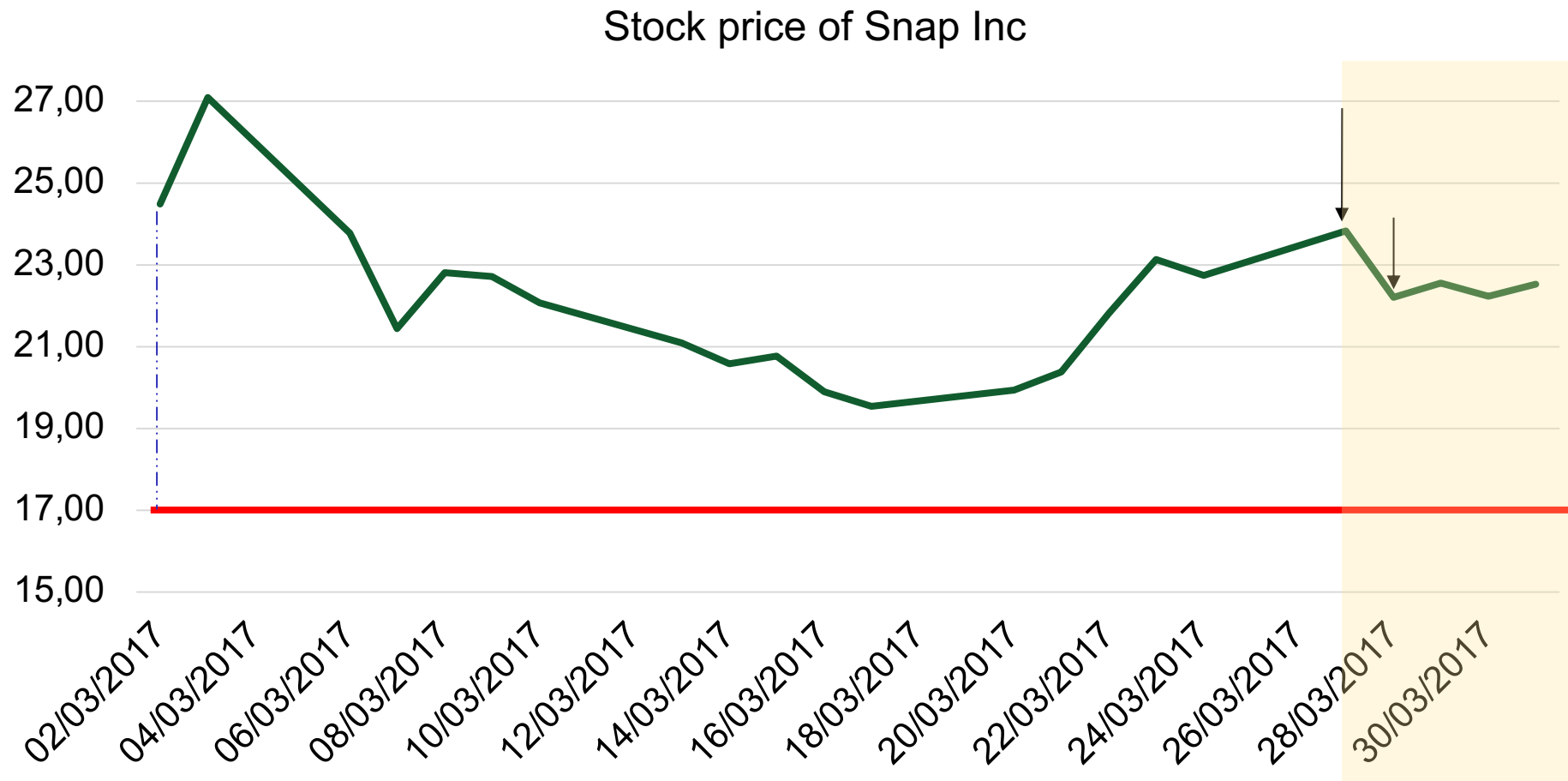
Snap, Inc.

2. IPO: Market sentiment during the IPO quiet period



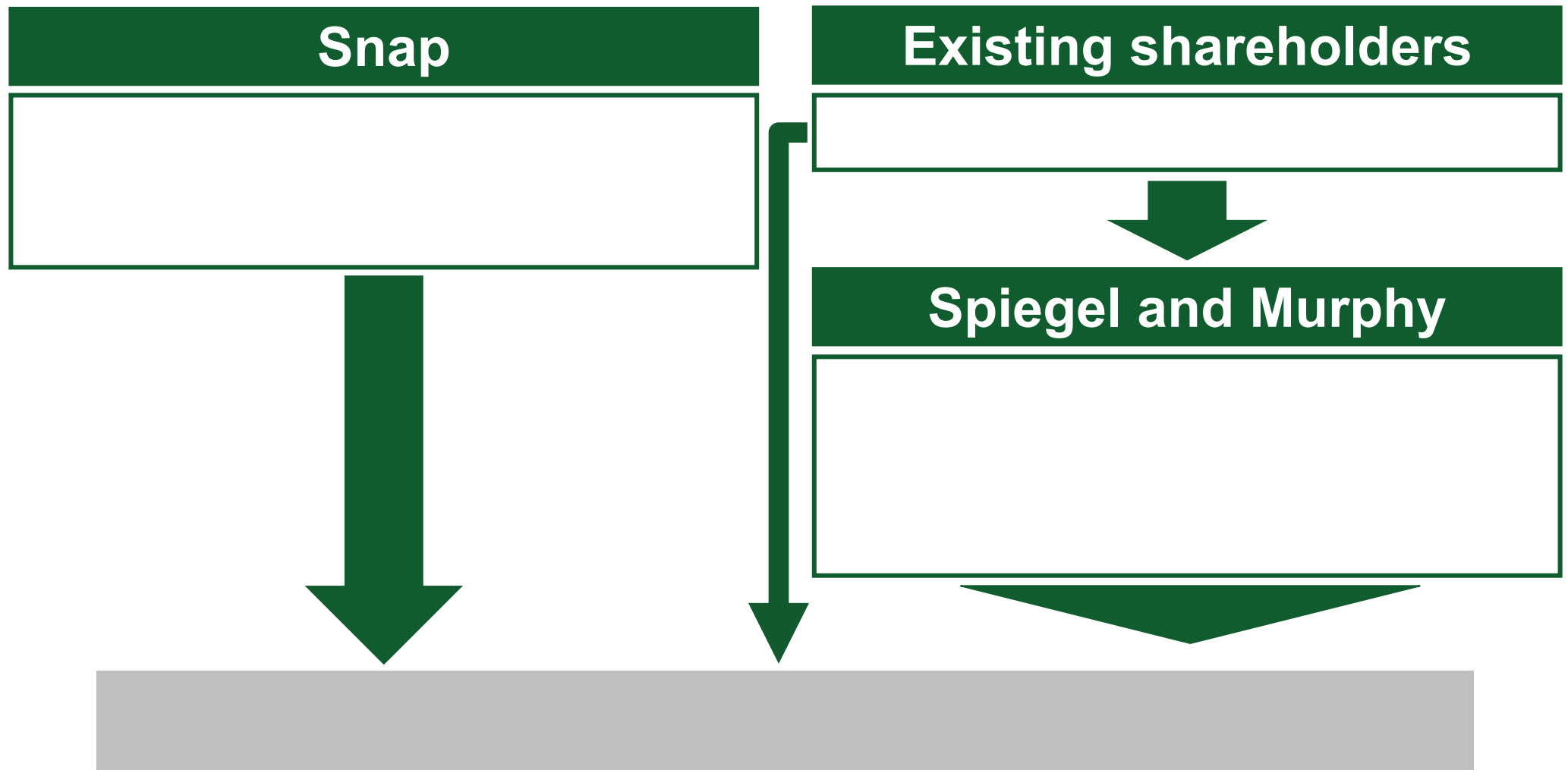
Source: Datastream

2. IPO: Market sentiment after the IPO quiet period



Source: Datastream

2. IPO: How does the share classification look like?



2. IPO: Characteristics of Snap's IPO

Underwriters

**Underwriter
fees**

**Stated purpose
of the offering**

2. IPO: The underpricing anomaly

2a. What is underpricing?

2b. Was Snap's stock underpriced?

2. IPO: Money “left on the table”

2c. How much money was “left on the table”?

2. IPO: Greenshoe

2d. What is a “greenshoe” option?

2e. Why was the greenshoe exercised?

2. IPO: Underwriters' performance

2f. Did the underwriters of the IPO do a good job?

2g. What could explain Snap's underpricing?

2. IPO: Underwriting fees

2h. How much did Morgan Stanley earn from Snap's IPO?

3. Kemp's challenge: Setting the stage

- Now let's take the position of Elizabeth Kemp. She is the portfolio manager of a \$400 million long-only, technology fund at Sand Hill Road Capital, and bought 500,000 Snap shares at the IPO.
- On Monday, March 27, 2017, the quiet period for Snap's IPO has just expired and Kemp is sitting at the desk in Palo Alto, California.
- She is reviewing the series of new analyst reports, most of which suggest buy recommendations on Snap. Before the market opens, she tries to find value relevant information from the reports.
- As a relatively young portfolio manager with a good 3-year track record, Kemp wondered whether now was the time to harvest her gain or double down and buy more shares.

3. Kemp's challenge: Buying or selling Snap shares?

BUY Shares

SELL Shares

3. Kemp's challenge: 4 portfolio-related facts

Amount

Return

Weight in portfolio

Past decisions

3. Kemp's challenge: What should she do?

Should she buy further shares or harvest her gains?

4. Valuation of Snap: Revenue forecast

Snap Advertising Revenue Model	2016	2020E	Notes
Daily Average Users (DAU, in millions)	158	236	User Base grows at 11% CAGR
X Hours per DAU per Day	0.42	0.50	
X Ads per DAU per Hour	0.6	7.9	Snap continues to grow ad load
X Average Pricing (\$ per 1K impressions)	\$26.00	\$14.37	Inventory scales and pricing falls
X 365 Days per Year	365	365	
Advertising Revenue (\$ mil)	\$378	\$4,889	Ad revenue grows at a 87% CAGR
Avg. Revenue per DAU (ARPU)	\$2.39	\$20.72	

Source: Exhibit 2

4. Valuation of Snap: Revenue vs. Comps

2016 Advertising Revenue Drivers	Snap	Instagram	Facebook	Twitter
Daily Average Users (DAU, in millions)	158	345	1,227	129
X Hours per DAU per Day	0.42	0.36	0.58	0.15
X Ads per DAU per Hour	0.6	6.6	48.6	104.3
X Average Pricing (\$ per 1K impressions)	\$26	\$6	\$2	\$3
X 365 Days per Year	365	365	365	365
Advertising Revenue (\$ mil)	\$378	\$1,795	\$25,118	\$2,254
Avg. Revenue per DAU (ARPU)	\$2.39	\$5.20	\$20.47	\$17.47

Source: Exhibit 3

4. Valuation of Snap: Financial forecast

(\$ millions)	2016Act	2017	2018	2019	2020	2021	2022	2023	2024	2025	TV Year (2026)
Net Revenue	\$405	\$944	\$1,935	\$3,254	\$4,902	\$6,907	\$9,189	\$11,635	\$14,131	\$16,569	\$17,149
Revenue Growth	589.5%	133.3%	105.0%	68.2%	50.6%	40.9%	33.0%	26.6%	21.4%	17.3%	3.5%
Adj. EBITDA*	(\$459)	(\$772)	(\$610)	(\$195)	\$363	\$1,474	\$2,616	\$3,888	\$5,223	\$6,574	\$6,804
Adj. EBITDA Margin	(113.6%)	(81.8%)	(31.5%)	(6.0%)	7.4%	21.3%	28.5%	33.4%	37.0%	39.7%	39.7%
- Cash Taxes	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$823	\$1,448	
Incr (Decr) in Net WC	\$151	\$257	\$228	\$231	\$228	\$209	\$224	\$248	\$207	\$151	
Capital Expenditures	\$67	\$83	\$97	\$109	\$120	\$130	\$140	\$148	\$155	\$161	
- Stock-based Comp.	\$32	\$75	\$159	\$203	\$257	\$317	\$421	\$533	\$648	\$759	
Free Cash Flow (FCF)	(\$709)	(\$1,186)	(\$1,093)	(\$737)	(\$243)	\$818	\$1,831	\$2,958	\$3,390	\$4,054	\$4,196

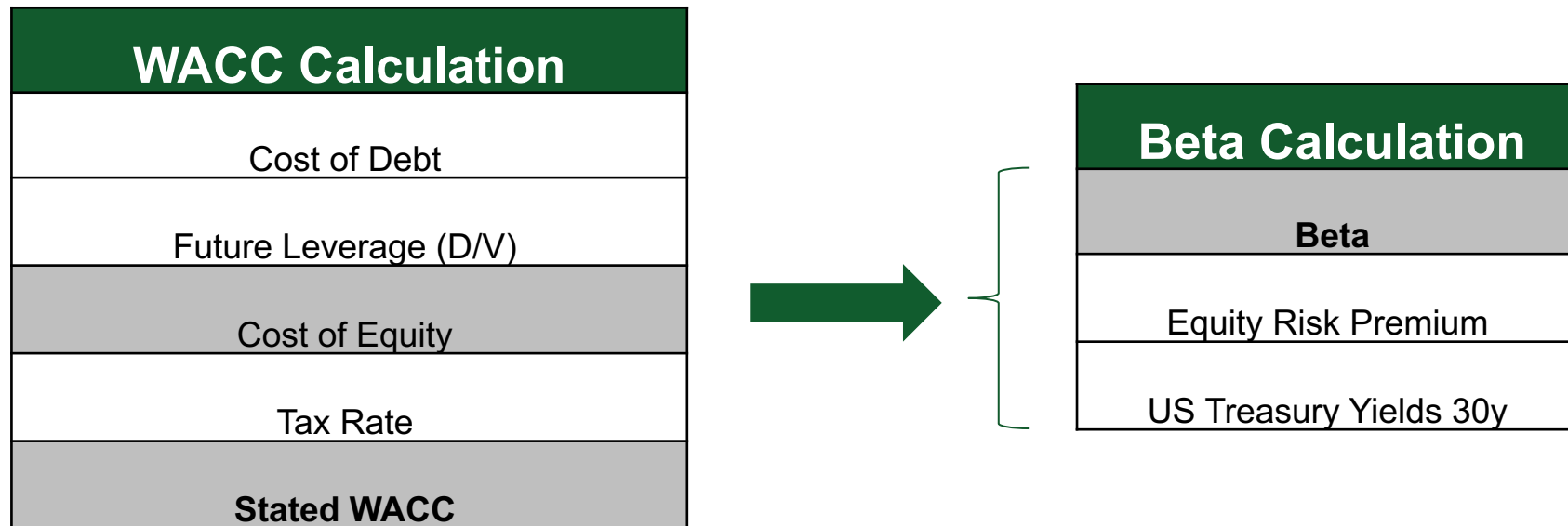
Source: Exhibit 10

* Adjusted EBITDA is a non-GAAP item which excludes stock-based compensation expense (a cash expense), but includes depreciation and amortization.

4. Valuation of Snap: Financial forecast (II)

- Analysts use a long forecast period to accommodate very high growth rates in the early years and allow sufficient time for growth to slow down in later years; the cash flows must be in a steady state before calculating a terminal value.
- Even with a 10-year forecast period, there is a dramatic decline in the revenue growth rate from 17.3% in 2025 to 3.5% in 2026.
- Snap does not pay taxes until 2024; the changing tax rate is another reason why the cash flows in the terminal year may not be in a steady state.
- Snap is “asset light”: CapEx and working capital are modest given that Snap’s sales grow from \$1 in 2017 to \$17 billion in 2026 as Snap uses third-party infrastructure partners (Google Cloud) to host its services and to avoid significant capital expenditures to support revenue generating activities.

4. Valuation of Snap: WACC calculation



WACC =

Cost of Equity = Risk-free rate + Beta · Market Risk Premium
→ Calculated implied equity Beta =



4. Valuation of Snap: Firm value and per share value

(\$ millions)	2016Act	2017	2018	2019	2020	2021	2022	2023	2024	2025	TV Year (2026)
Free Cash Flow (FCF)	(\$709)	(\$1,186)	(\$1,093)	(\$737)	(\$243)	\$818	\$1,831	\$2,958	\$3,390	\$4,054	\$4,196
Terminal Value Growth = 3.5%											
Present Value (FCF+TV) @ 9.7%											
Total Value											
+ Cash	\$2,081										
- Debt	\$0										
Equity Value											
# shares (millions)	1,404										
Per share value											



Source: Exhibit 10

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4. Valuation of Snap: Brian Nowak

Education

- BBA in Accounting and Finance from the University of Michigan
- Master of Accounting from the University of Michigan



Work Experience

- Associate (Financial Advisory Services) at PricewaterhouseCoopers (PwC)
- Equity Research Associate at Sanford C. Bernstein
- VP at Nomura Securities covering US internet stocks
- Internet Equity Research Analyst at Susquehanna International Group
- Managing Director in charge of the US Internet Research Team at Morgan Stanley

4. Valuation of Snap: Brian Nowak (II)

Awards

- Runner Up in the 2016 Institutional Investor poll
- Holds the Chartered Financial Analyst (CFA) designation
- According to TipRanks (a web site that ranks equity analysts), Nowak:
 - Was ranked #296 out of 4,826 analysts as of June 7, 2018 (top 10%)
 - Has a 73% success rate (% of recommendations that were profitable over 1 year)
 - Has an average return of 12.8% based on his buy and sell recommendations

4. Valuation of Snap: Brian Nowak's price target of \$28

4a. How did Nowak get a price target of \$28 per share if Morgan Stanley's DCF valuation should generate only \$25 per share?

4. Valuation of Snap: Key inputs – WACC

4b. Is the WACC of 9.74% relevant to Nowak's valuation?

4. Valuation of Snap: Key inputs – Growth rate

4c. Is the terminal value growth rate (TVGR) of 3.5% relevant to Nowak's valuation?

4. Valuation of Snap: Key inputs – Forecast

4d. Are the FCF forecast and Snap's 2020 revenue forecast relevant to Nowak's valuation?

5. Conflicts of interest: Stanley vs. Fitzgerald

- The 2 analysts, Brian Nowak from Morgan Stanley and Kip Paulson from Cantor Fitzgerald are fairly representative of the 30 analysts that covered Snap when the IPO quiet period expired; focusing on them helps elicit the “informed vs. conflicted” debate.
- Nowak’s \$28 valuation and overweight (buy) recommendation are at the bullish end of the spectrum; Paulson’s \$18 price target and underweight (sell) recommendation are at the bearish end of the spectrum.
- To what extent should Kemp’s decision be guided by Nowak’s or Paulson’s opinions?

5. Conflicts of interest: With clients

5a. Is Nowak more informed or more conflicted regarding Morgan Stanley's investment banking clients?

5. Conflicts of interest: With the underwriter (bank)

5b. Is Nowak, as an affiliated analyst, more informed or more conflicted regarding Morgan Stanley itself?

5. Conflicts of interest: Code of Conduct and Disclosure

5c. Are codes of conduct and conflicts of interest disclosures effective to prevent conflicts?

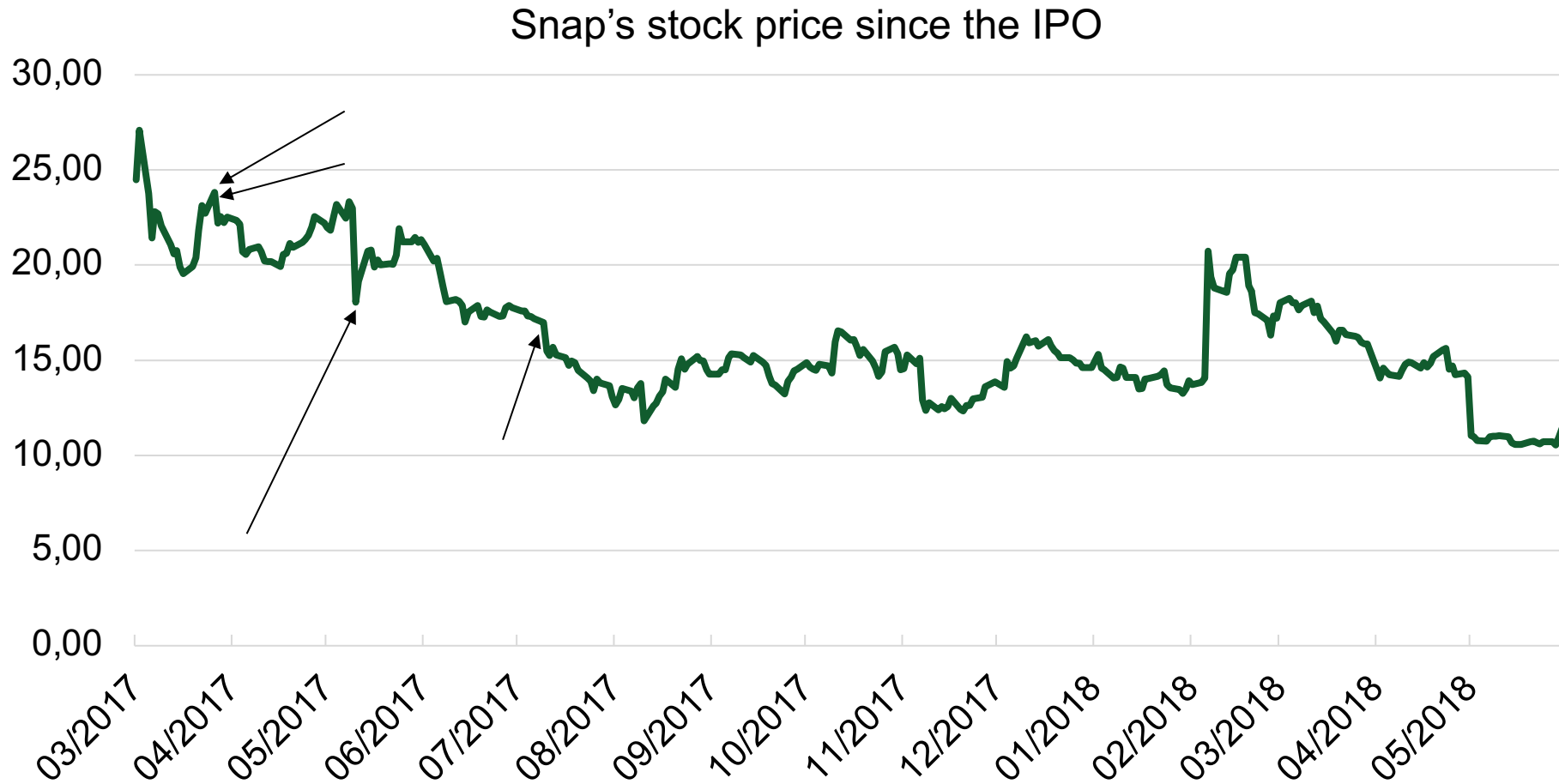
5. Conflicts of interest: Affiliated vs. unaffiliated analysts

- There is a sharp divergence between the unaffiliated analysts who wrote reports during the quiet period and the affiliated analysts who wrote reports after the quiet period expired:
 - The unaffiliated analysts were much more pessimistic than the affiliated ones
 - 43% of unaffiliated analysts had sell recommendations on Snap, none of the affiliated had a sell recommendation
 - 63% of the affiliated had a buy recommendation vs. only 14% for the unaffiliated

5. Conflicts of interest: Research vs. investment banking

- Conflicts between equity research and investment banking may also exist:
 - Across the entire universe of covered firms, buy ratings are 1.8 times more prevalent than sells
 - In the sub-sample of covered firms that are investment banking clients of the bank, buy ratings are 3.8 times more prevalent than sells
 - While it is possible investment banking clients happen to be better firms with significant upside potential, it is also possible this distribution reflects a positive bias in favor of clients

6. Excursus: What is Kemp's final decision?



Source: Datastream

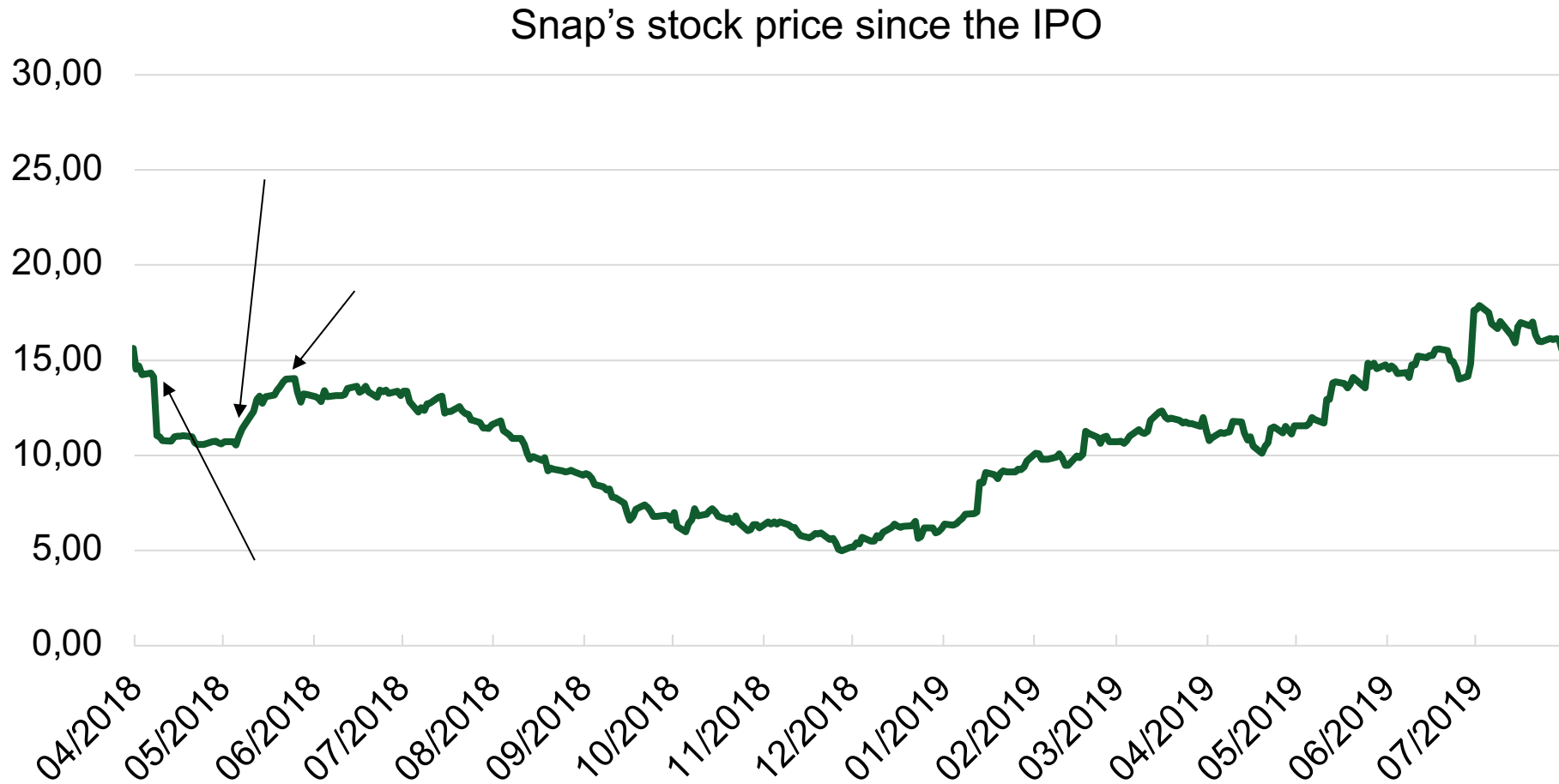
6. Excursus: Nowak's revised report (March 28, 2017)

- First, in response to the flurry of new analyst reports, Snap's price jumped almost 5% on Monday, March 27.
- Nowak discovered an error in his valuation model that overstated his FCF forecast. Because the 2 changes exactly offset each other, lower FCF and lower WACC, the calculated NPV and price target of \$28 remained the same.
- Fixing the "tax calculation error" caused forecasted FCF to fall by 30% to 40% in the later years but Nowak also reduced the WACC from 9.74% to 8.00% to make it consistent with other companies he covered. He reduced the WACC to reflect a reduction in the equity risk premium from 5.58% to 4.29%. Nowak had done the same thing, used offsetting corrections to the FCFs and the WACC to generate a constant price target on at least one other occasion (Amazon).
- Nowak only acknowledged the error without explaining the nature of the error although all of the difference between the original and the revised FCF figures was due to a change in the adjusted EBITDA line. Also, it is unclear why a tax calculation error would affect EBITDA as adjusted EBITDA is a pre-tax number.

6. Excursus: Nowak's re-revised report (July 11, 2017)

- After Snap reported worse than expected results in its first quarter as a public company which caused its stock price to fall by 21%, Nowak admitted that he had been wrong about Snap in a new report published on July 11, 2017.
- Nowak claimed that he overstated Snap's ability to innovate and improve its ad product in 2017 and user monetization. He believes that Instagram has become more aggressive in competing for Snap's ad dollars. He sees Snap revenue growth lower than the previously expected causing the DCF-based price target to fall by 40% from \$28 to \$16.
- Although we talked about underpricing earlier, Snap's underperformance is consistent with a second IPO anomaly: long-term underperformance; Snap's stock price fell below \$5 in December 2018 and was around \$15 on July 11, 2019, 2 years after Nowak's "mea culpa".
- According to Ritter (2019), the average 3-year buy-and-hold return for IPOs from 1980-2017 was 21.9% but was -17.9% on a market-adjusted and -7.1% on a style-adjusted basis relative to matched firms (See lecture 5, slide 41-42).

6. Excursus: What happened since the IPO?



Source: Datastream

Recap

- The decision to invest in a stock depends in large part on beliefs about how much this share is actually worth.
- Key inputs such as the FCF forecast and discount rate (WACC), play an important role in DCF valuation.
- Recommendations to buy or sell a stock can diverge among research analysts from different financial institutions.
- Research analysts from (co-)lead underwriters of an IPO may be more informed about a stock and be more conflicted. However, codes of conduct should mitigate such conflicts.

Mandatory readings

- Di Maggio, M., and B.C. Esty, 2018, Valuing Snap After the IPO Quiet Period (A, B&C), Harvard Business School, Case 218-095, 218-096 and 218-116.

Additional readings

- Brown, L.D., Call, A.C., Clement, M.B., and N.Y. Sharp, 2015, Inside the «black box» of sell-side financial analysts, Journal of Accounting Research, 53,1-47.
- Ritter, J.R., 2018, Money left on the table in IPOs by firm, Working paper, University of Florida.
- Ritter, J.R., 2019, Initial public offerings: Updated statistics on long-run performance, Working paper, University of Florida.
- Soltes, E., 2014, Private interaction between firm management and sell-side analysts, Journal of Accounting Research 42, 245-272.